Are UFO Sightings Explainable Phenomena?

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What are 'Unidentified Aerial Phenomena' (UAPs)?

Research Questions

- What is the incidence of UAP reports over time, and how do they vary geographically?

- Does the geographical distribution of UAPs correlate with quantifiable data that is publicly available:
  - Commercial space launch sites;
  - Military base locations; and/or
  - Commercial airport locations?

- What are reported shapes of UAPs, and does this vary by year or states?
- What is the duration of reported UAP events, and has this changed over the years?

Chosen Datasets

- <u>Primary</u>: NUFORC (National UFO Research Center)
  - Consists of human-recorded unidentified aerial phenomena (UAP) events
  - *Key Fields*: City, state, time, date, duration, shape
- <u>Secondary</u>:
  - Commercial Space Launches
    - *Key Fields:* Date, state of launch
  - Known Military Bases
    - *Key Fields:* City
  - Airport Locations
    - *Key Fields:* City

Data Cleaning

NUFORC (Before)

	Event_Time	Event_Date	Year	Month	Day	Hour	Minute	City	State	Shape	Duration	Summary	Event_URL
0	2017-04-20T14:15...	2017-04-20	2017.0	4.0	20.0	14.0	15.0	Palmyra	NJ	Other	5 minutes	I observed an ai...	http://www.nufor...
1	2017-04-20T04:56...	2017-04-20	2017.0	4.0	20.0	4.0	56.0	Bridgeview	IL	Light	20 seconds	Bridgeview, IL, ...	http://www.nufor...
2	2017-04-19T23:55...	2017-04-19	2017.0	4.0	19.0	23.0	55.0	Newton	AL	Triangle	5 seconds	Silent triangle ...	http://www.nufor...
3	2017-04-19T23:50...	2017-04-19	2017.0	4.0	19.0	23.0	50.0	Newton	AL	Triangle	5-6 minutes	My friend and I ...	http://www.nufor...
4	2017-04-19T23:29...	2017-04-19	2017.0	4.0	19.0	23.0	29.0	Denver	CO	Light	1 hour	Moved slow but m...	http://www.nufor...

NUFORC (After)

City	UFO_State	Shape	Duration	Summary	Event_URL	Date	Converted_Duration_min	Date_Clean	Year	City_State	Region	Sub-Region	AM_PM
Palmyra	NJ	Other	5 minutes	I observed an ai...	http://www.nufor...	04/20/17	5.000000	2017-04-20	2017	Palmyra, NJ	Northeast	Mid Atlantic	P.M.
Bridgeview	IL	Light	20 seconds	Bridgeview, IL, ...	http://www.nufor...	04/20/17	0.333333	2017-04-20	2017	Bridgeview, IL	Midwest	East North Central	A.M.
Newton	AL	Triangle	5 seconds	Silent triangle ...	http://www.nufor...	04/19/17	0.083333	2017-04-19	2017	Newton, AL	South	East South Central	P.M.
Newton	AL	Triangle	5-6 minutes	My friend and I ...	http://www.nufor...	04/19/17	0.000000	2017-04-19	2017	Newton, AL	South	East South Central	P.M.
Denver	CO	Light	1 hour	Moved slow but m...	http://www.nufor...	04/19/17	60.000000	2017-04-19	2017	Denver, CO	West	Mountain	P.M.

Assumptions

- NUFORC
  - Focused on events beginning in 1974 (when NUFORC was created)
  - Included events that occurred in the 50 U.S. states & D.C.
- Space Launches
  - Removed non-U.S. launches
- Military Bases
  - Kept all U.S. and corresponding territory (i.e. Guam, Puerto Rico) bases
- Airports
  - Kept airports only in the U.S. or its territories

NUFORC Events By State & City

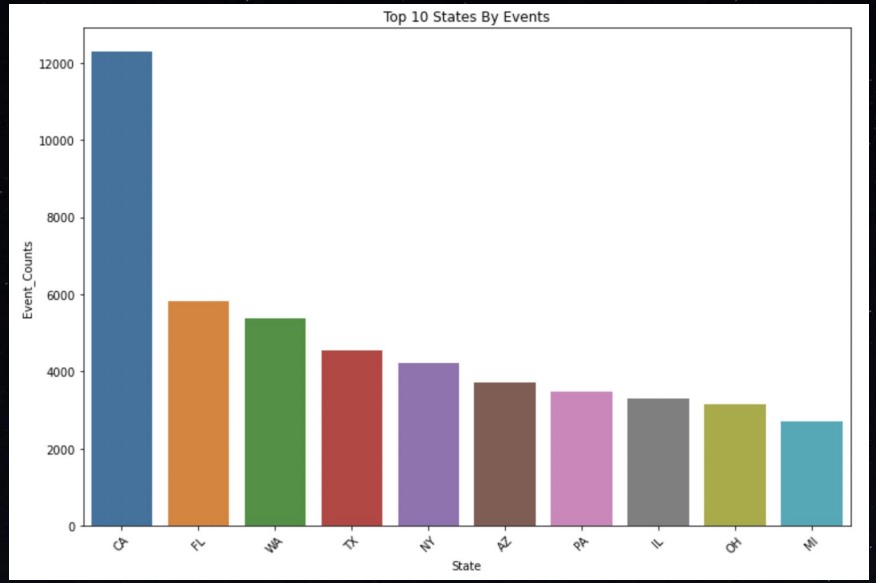

Figure A: Top 10 Ranking States for Sightings

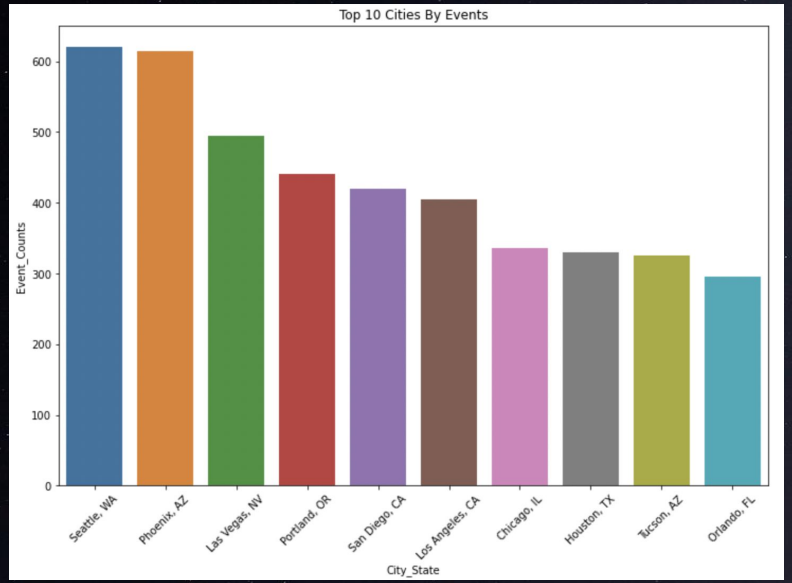

Figure B: Top 10 Ranking Cities for Sightings

NUFORC Events By Decade/Year

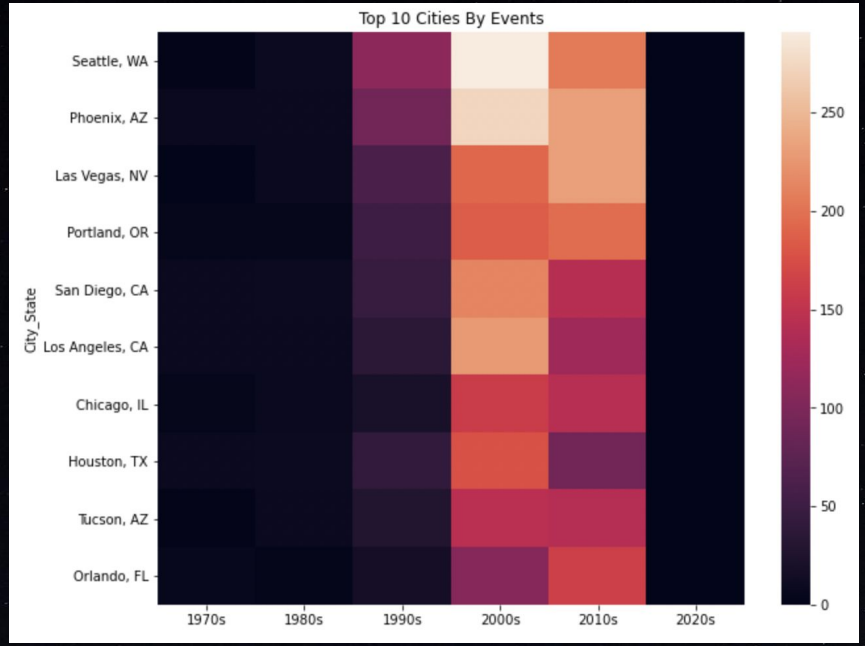

Figure C: Top 10 Ranking Cities for Sightings By Decade

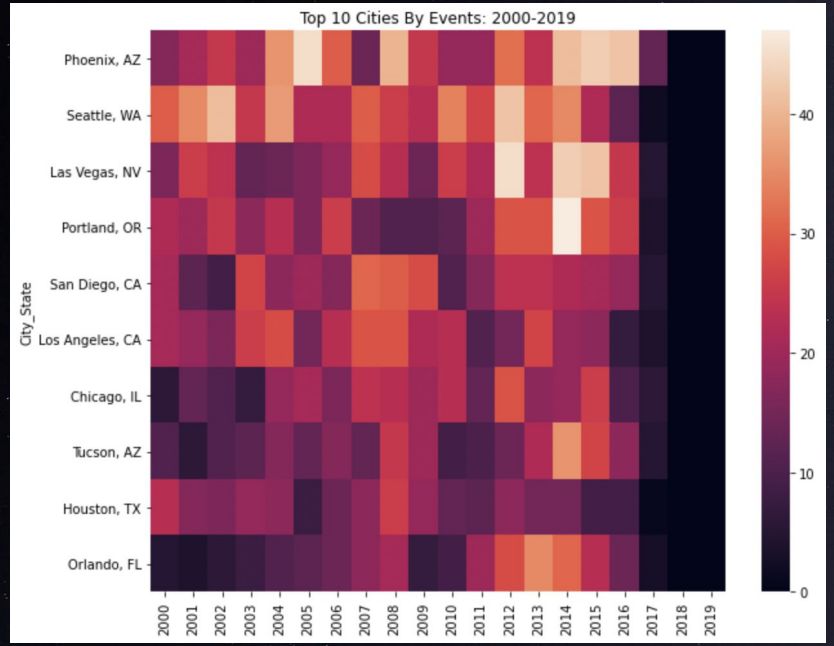

Figure D: Top 10 Ranking Cities for Sightings (2000-2019)

Shapes & Reported Times of UAPs

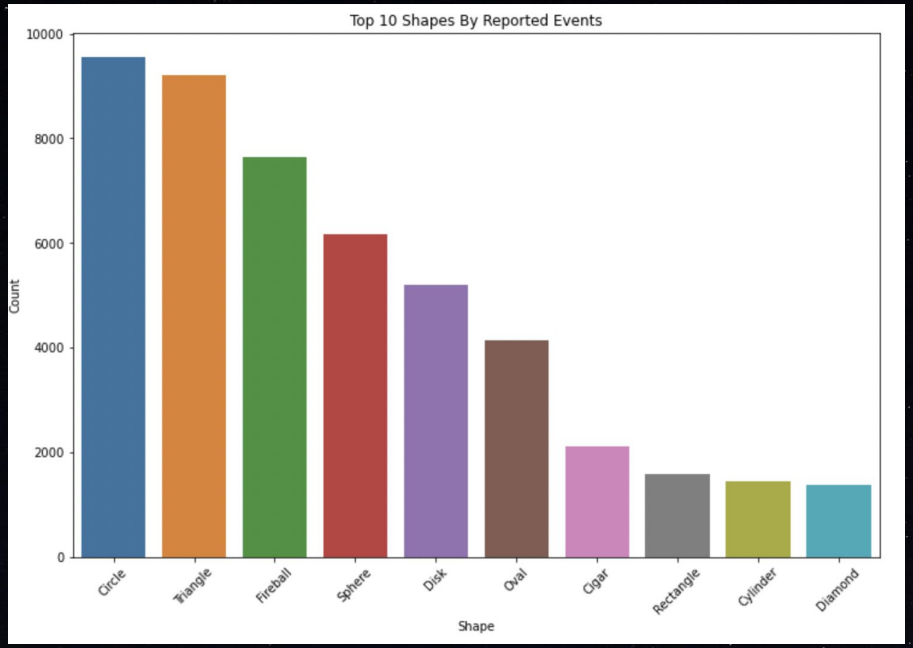

Figure F: Top 10 Shapes of UAP Events

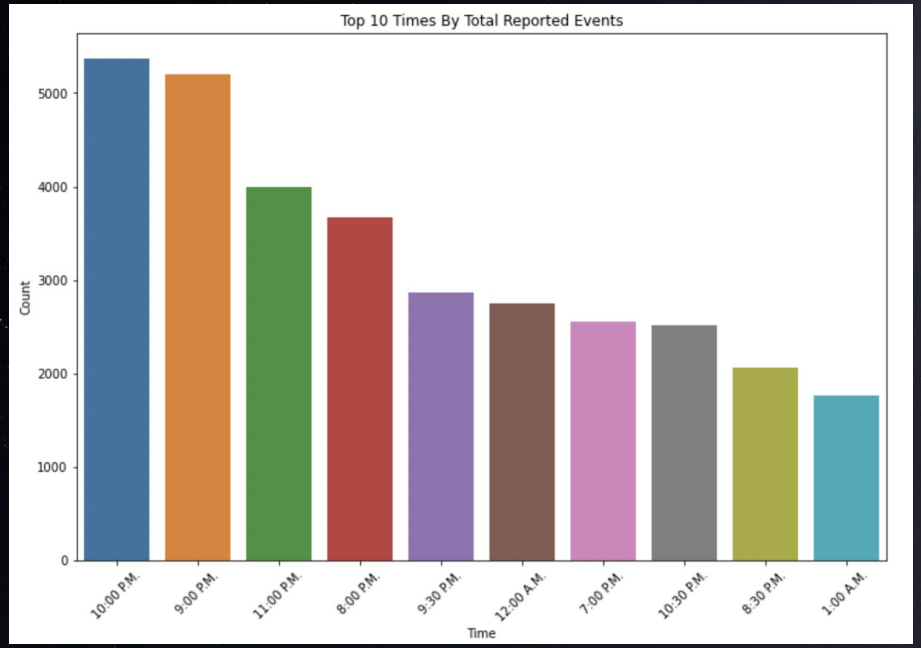

Figure G: Top 10 Times of UAP Events

Comparing UAP Sightings With Secondary Datasets

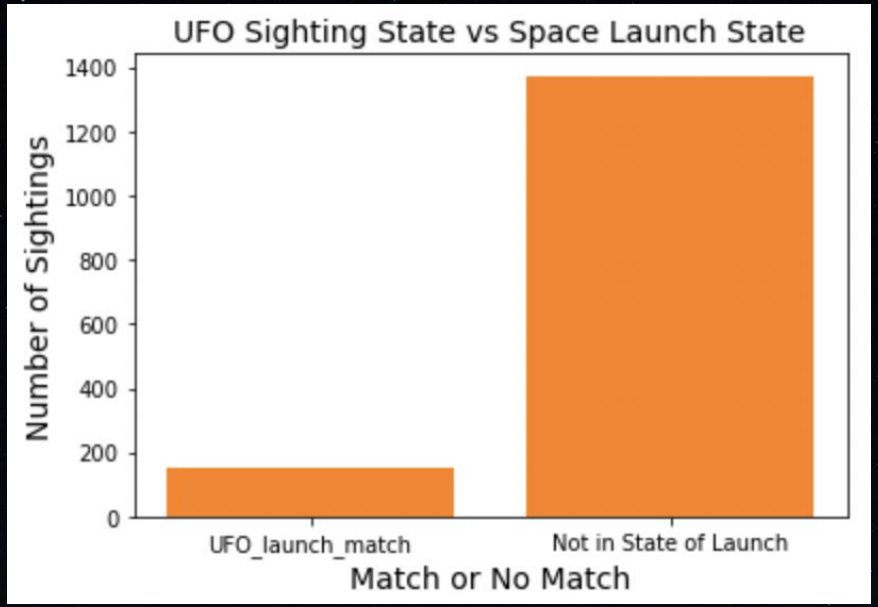

Figure E: Comparing States of UAP Events versus States of Space Launches

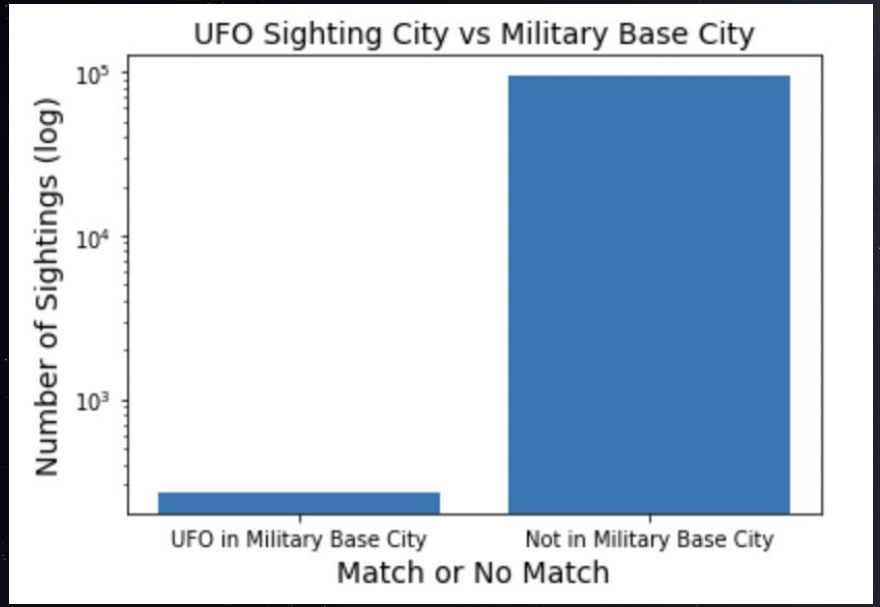

Figure F: Comparing Cities of UAP Events versus Cities of Military Bases

Reported Duration of UAPs

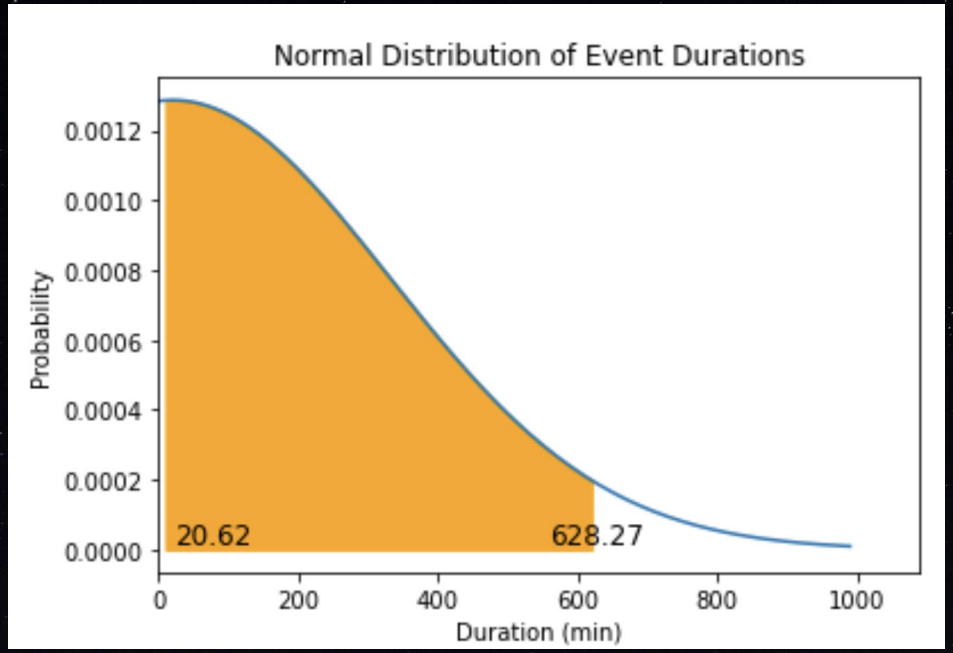

Figure I: Normal Distribution of Event Durations

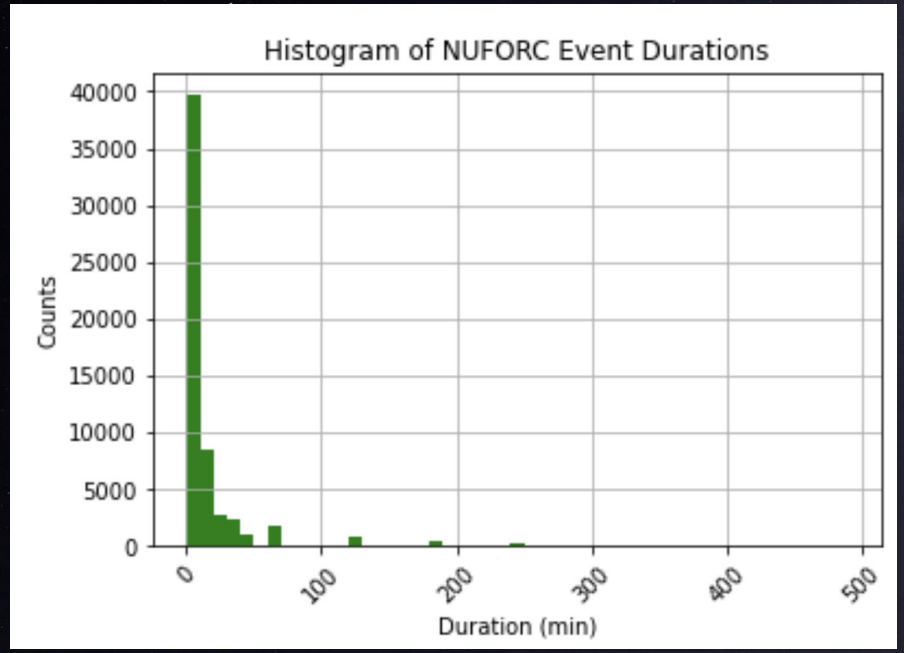

Figure J: Histogram of Event Durations

Conclusions

- Our exploratory data analysis reveals no clear association between the frequency of UFO sightings and either locations and dates of space launches, or locations of military bases, or airports.
- On EDA, we were unable to find any clear link between UFO sightings and natural phenomena such as aurora borealis.
- Thus, from our EDA the majority of UFO remain unidentified!
- The top states for sightings are California and Florida.
- The number of UFO sightings has increased year over year since 1970, particularly from 2000 to 2019.
- Most sightings occurred late in the evening and lasted on average ~20.5 min.

Bibliography

Dataset Sources:

- **Primary Dataset:** NUFORC (https://data.world/khturner/national-ufo-reporting-center-reports/workspace/file?filename=nuforc_events.csv)
- **Secondary Datasets**:
  - Historical Space Launches: https://www.faa.gov/data_research/commercial_space_data/launches/
  - Known Military Bases: https://public.opendatasoft.com/explore/dataset/military-bases/table/
  - Airport Locations: https://data.world/tylerudite/airports-airlines-and-routes